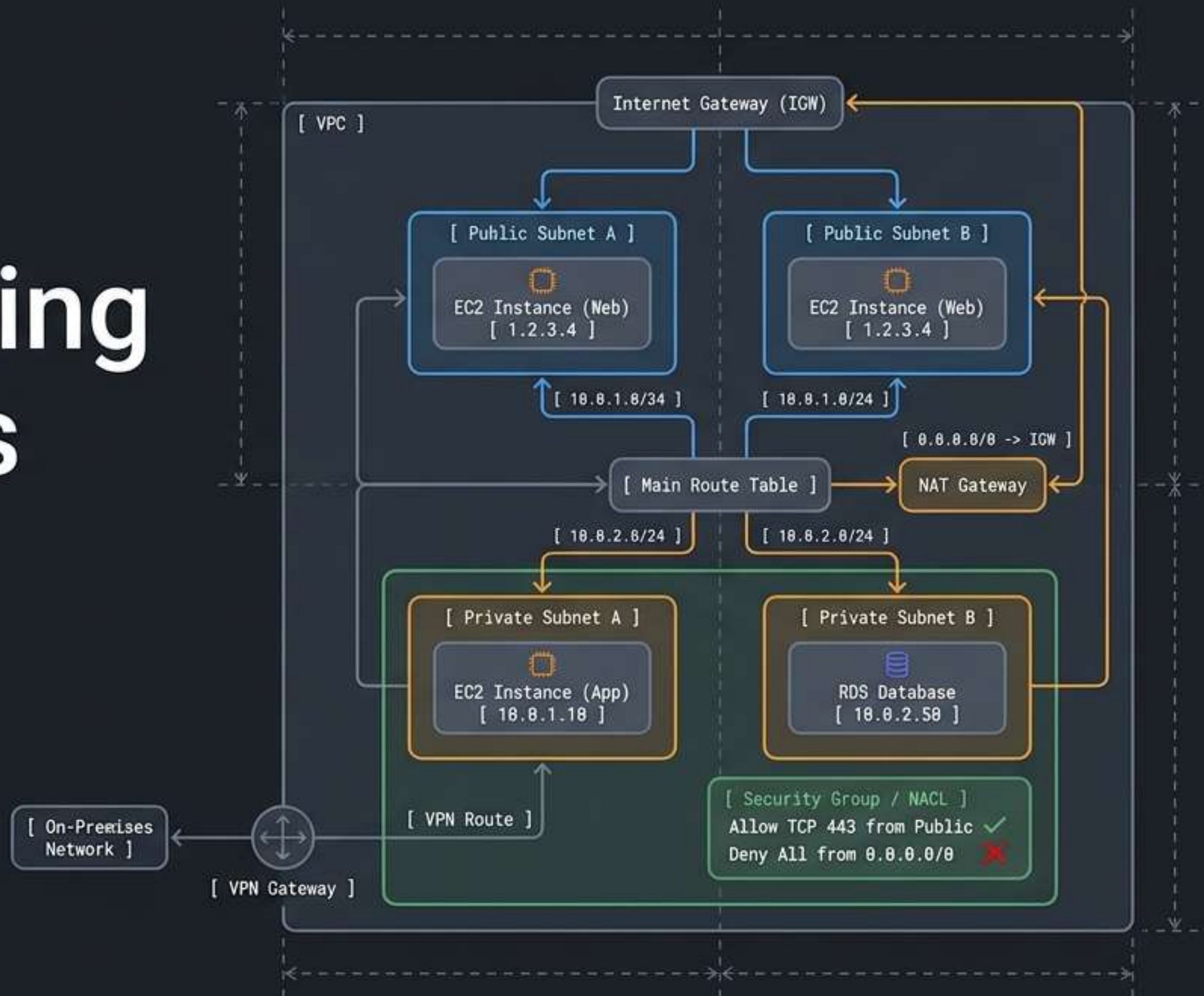


AWS Networking Fundamentals

VPC, Subnets, Routing, and Security Design



The Macro Container: Virtual Private Cloud (VPC)

A logical network boundary, not a physical data center.

What is it?

A software-defined, logically isolated virtual network inside AWS.

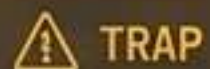
Why is it needed?

Provides the core environment to control IP addressing, routing, and access.

How is it sized?

Defined by a CIDR block (Classless Inter-Domain Routing).

Example: 10.0.0.0/16 (A large network space)



TRAP

Is a VPC a physical network?

No. It is a logical virtual network implemented on AWS's physical infrastructure.

[AWS Cloud]

[Region]

[VPC (CIDR: 10.0.0.0/16) - Your logical network]

Network Segmentation: Subnets & Availability Zones (AZs)

Dividing the address space for highly available architecture.

What is a Subnet?

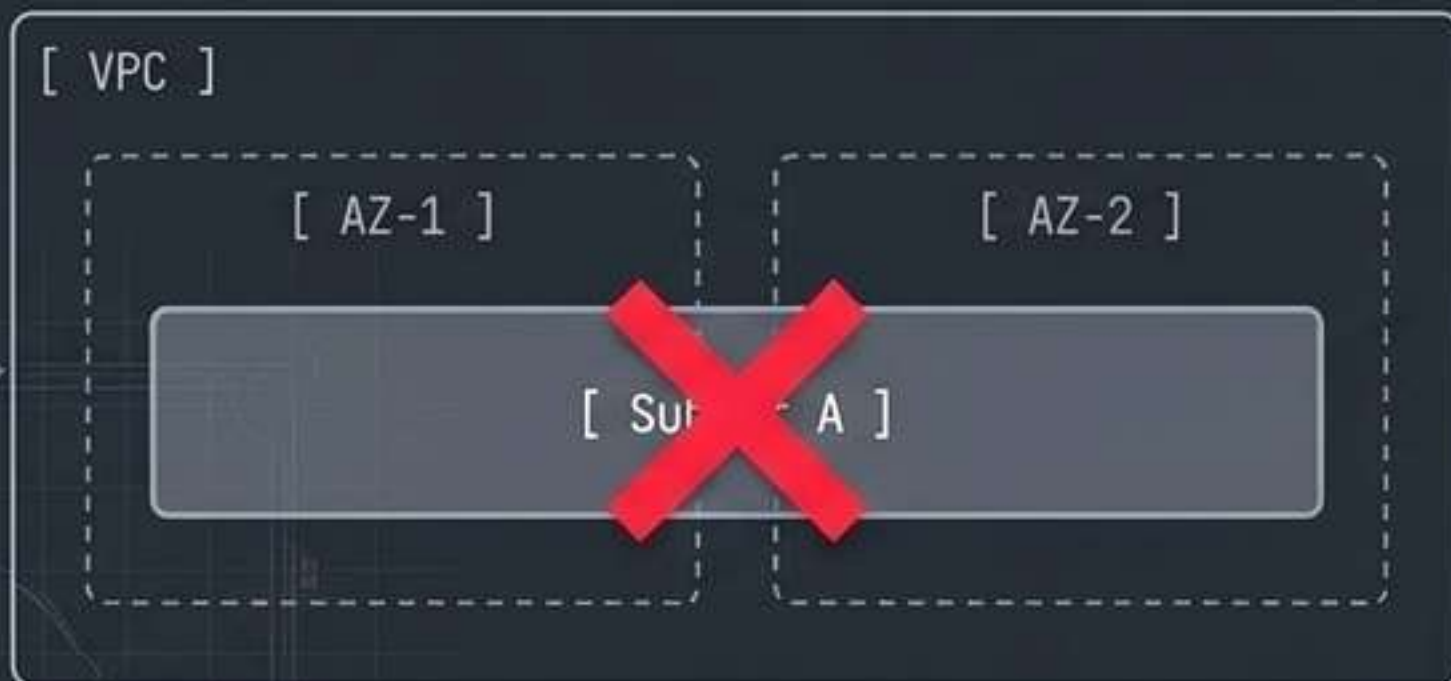
A smaller IP network carved out of your VPC (e.g., `10.0.1.0/24`).

Why divide?

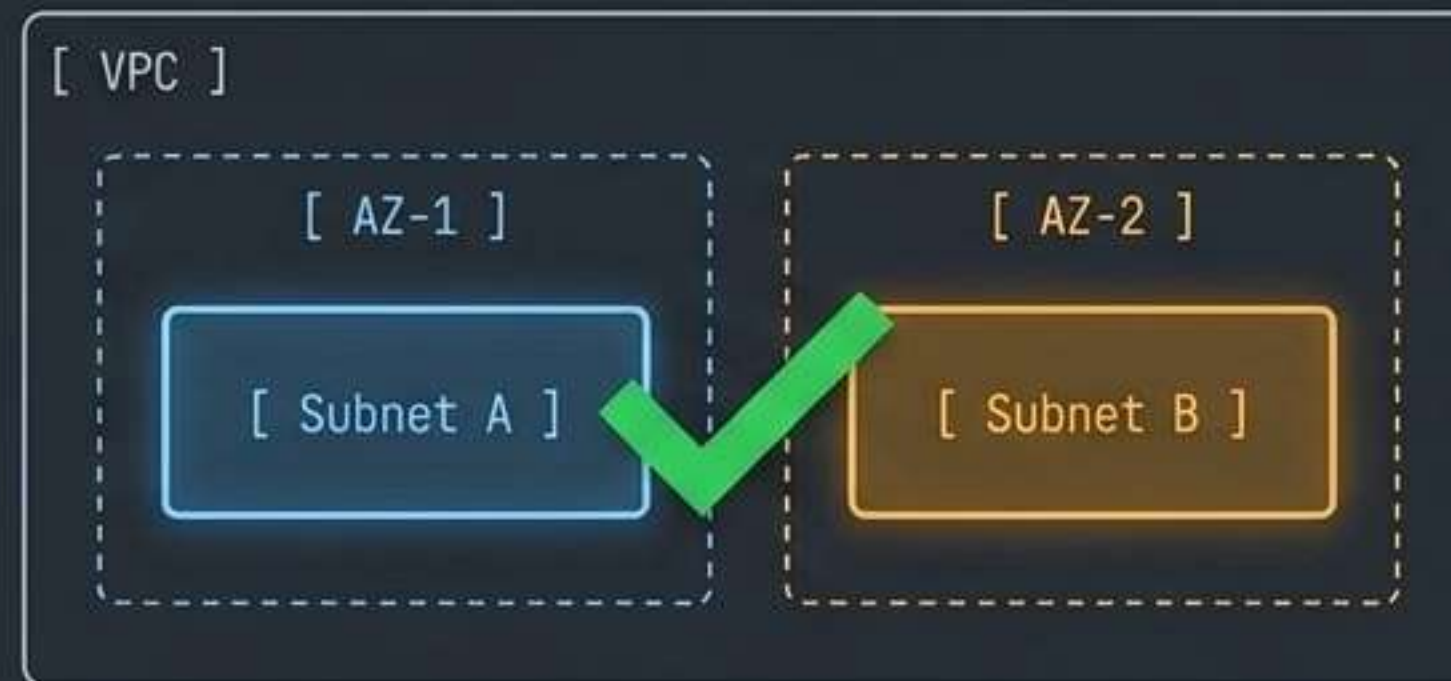
Different workloads (Databases vs. Load Balancers) require different security and routing rules.

💡 THE GOLDEN RULE OF AZs:

An AWS subnet is associated with ONE Availability Zone only. It cannot span multiple AZs.



The Wrong Way



The Right Way

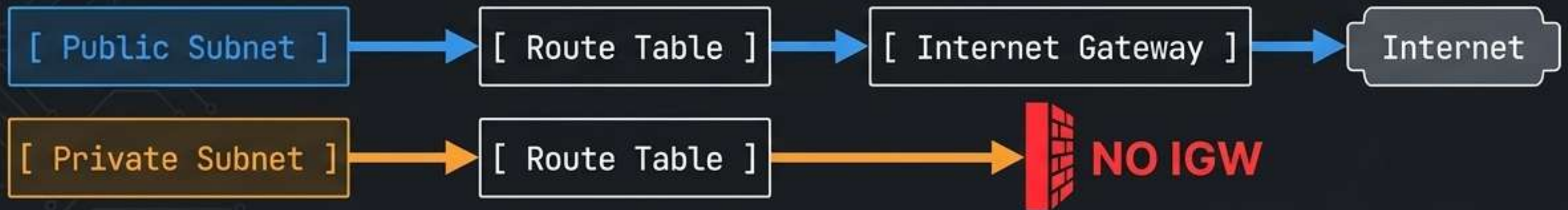
Public vs. Private Subnets

Routing determines visibility, not the subnet's name.

💡 MENTAL MODEL

A subnet is ONLY considered public if its route table has a direct route to an Internet Gateway (IGW).

Feature	Public Subnet	Private Subnet
Route to IGW	Yes	No
Direct Inbound Internet	Possible (with right configs)	Not directly
Typical Resources	Load Balancers, Web-facing	Databases, Backend Servers
Security Exposure	Higher (The Front Door)	Lower (The Internal Layer)



The Internet Gateway (IGW) Access Checklist

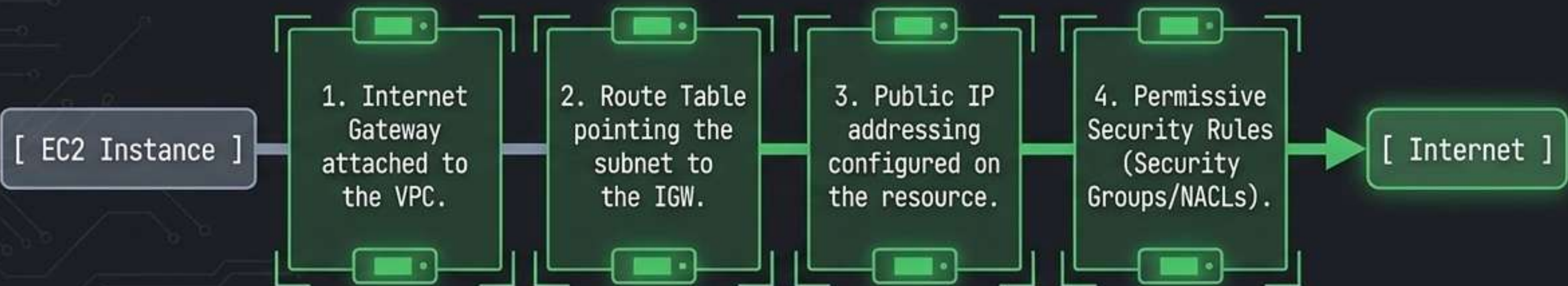
Connecting the VPC to the outside world requires multiple keys.

What is an IGW?

A VPC component that enables communication between resources and the internet.

⚠ CRITICAL TRAP: Having an Internet Gateway

Having an Internet Gateway does NOT automatically make every resource public. Internet access requires four distinct configurations.



NAT Gateway: Outbound Access for Private Subnets

Internet connectivity without inbound exposure.

Why is it needed?

Private resources (like backend servers) still need to download patches, access external APIs, or retrieve updates.

How does it work?

It allows private resources to initiate outbound connections to the internet, while blocking unsolicited inbound connections.

The Traffic Flow:

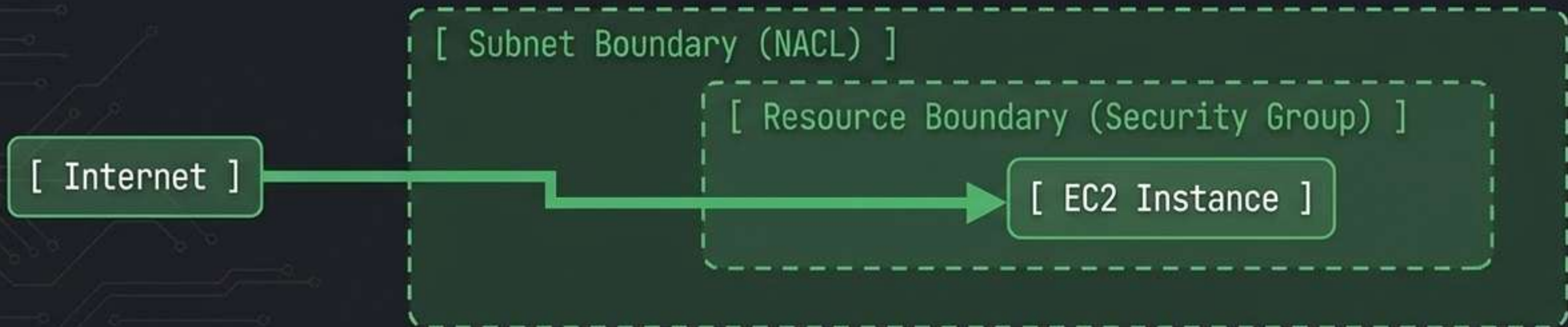
Private Server -> NAT Gateway -> Internet Gateway -> Internet



Security Layers: Security Groups vs. Network ACLs

Network segmentation reduces attack surface, but firewalls enforce the rules.

Security Group (SG)	Network ACL (NACL)
Scope: Resource-level (e.g., EC2, RDS)	Scope: Subnet-level
State: STATEFUL (Return traffic allowed automatically)	State: STATELESS (Return traffic must be explicitly allowed)
Rules: Allow rules only	Rules: Allow + Deny rules
Role: Virtual Firewall for the instance	Role: Network Firewall for the boundary



Synthesis: The Single-AZ Architecture

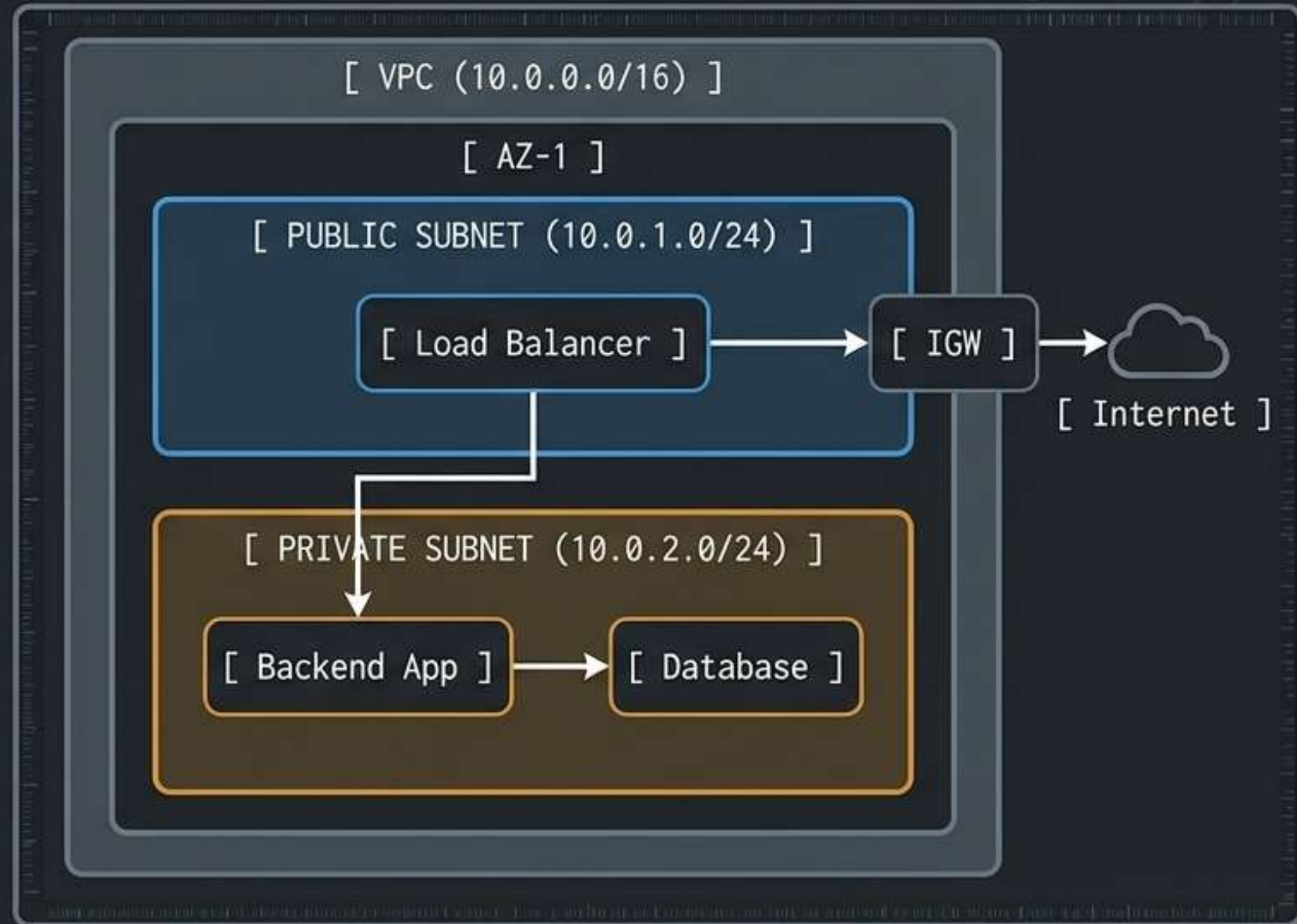
Visualizing the fundamental network blueprint.

The Principle of Least Exposure:

Databases should generally not be directly exposed to the public internet. Network segmentation creates distinct layers of trust.

Traffic Flow:

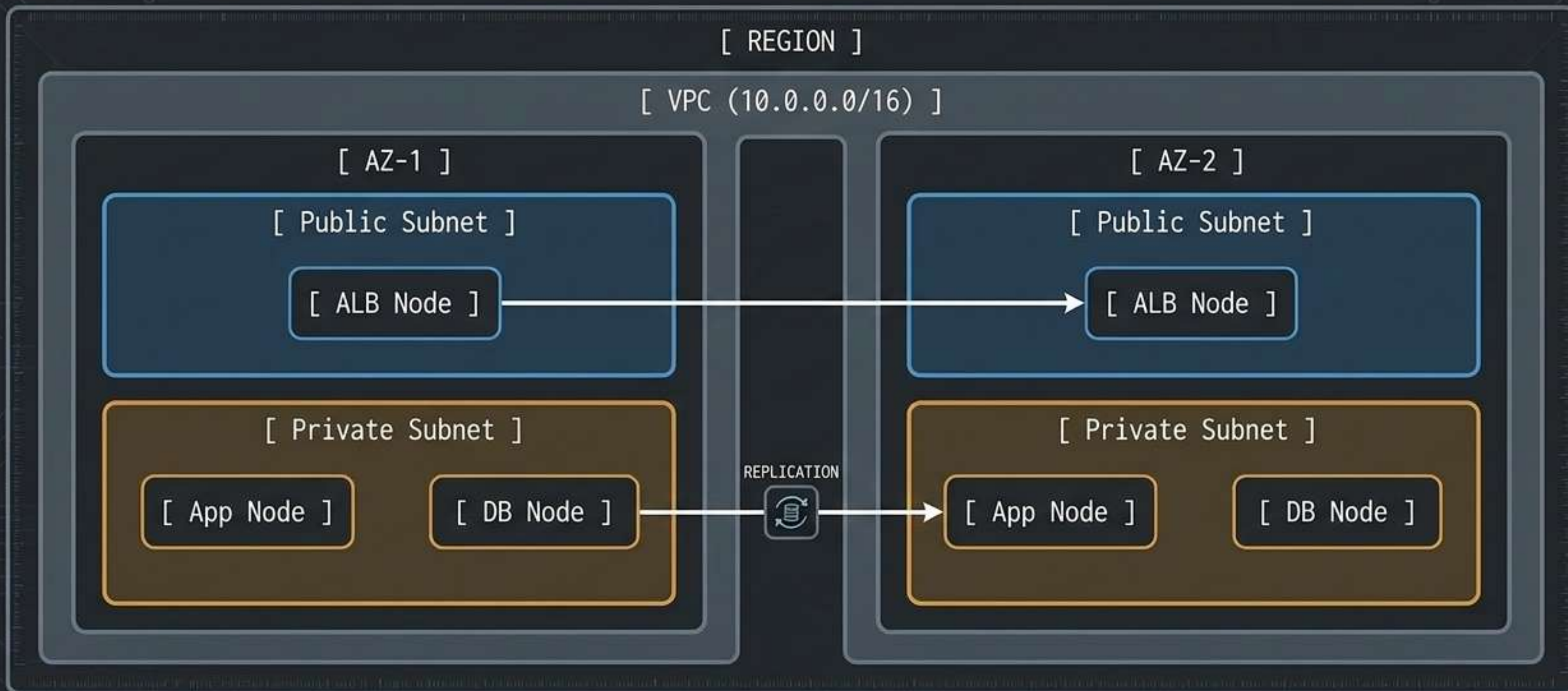
1. Internet hits IGW.
2. Traffic routes to Load Balancer in Public Subnet.
3. Load Balancer forwards to Backend in Private Subnet.
4. Backend queries Database in Private Subnet.



Synthesis: The Multi-AZ Production Architecture

The ultimate mental model: Region -> VPC -> AZ -> Subnet -> Resources.

Why Multi-AZ? High Availability. If an entire physical data center (AZ-1) goes down, the application continues running seamlessly from AZ-2.



Common Beginner Traps & Misconceptions

Fixing flawed intuition before it becomes a habit.



['A VPC is the same as a subnet.']

✗ Reality: A subnet is a smaller IP network carved out INSIDE a VPC.



[Reality: A subnet is a smaller IP network carved out INSIDE a VPC.]

✓ Reality: One Subnet = One AZ. No exceptions.



['A Private Subnet means the server is completely isolated from the internet.']

✗ Reality: It simply means no direct route to an IGW. It can still have outbound internet via a NAT Gateway.



[Reality: Security Groups. It requires SGs, NACLs, Route Tables, and

✓ Reality: Security is layered. It requires SGs, proper gateway configurations.

Quick Revision Cheat Sheet

Vocabulary mapped strictly to function

[VPC]	Logical isolated network boundary in AWS.
[CIDR]	Defines the total IP address space (e.g., 10.0.0.0/16).
[SUBNET]	Smaller network inside a VPC (Remember: 1 Subnet = 1 AZ).
[PUBLIC SUBNET]	Has a route table pointing to an Internet Gateway.
[PRIVATE SUBNET]	Has no direct route to an Internet Gateway.
[INTERNET GATEWAY (IGW)]	Connects the VPC to the internet.
[NAT GATEWAY]	Allows private resources outbound internet access.
[SECURITY GROUP]	Stateful resource-level firewall.
[NACL]	Stateless subnet-level traffic control.

The 3 Architectural Imperatives

If you remember nothing else, remember these principles.

1

The Structural Hierarchy

Region -> VPC -> Availability Zone -> Subnet -> Resource

This is the chain of containment for all AWS networking.

2

Routing Dictates Visibility

A subnet's name means nothing. A subnet is only "Public" if its route table explicitly points to an Internet Gateway.

3

The Principle of Least Exposure

Databases and backend services belong in Private Subnets. Security is achieved through strict network segmentation and defense-in-depth firewalls.

Mentor & Interview Questions

Test your conceptual understanding before moving on.

- 1 >_ Explain the strict relationship between an AWS Subnet and an Availability Zone.
- 2
- 2 >_ You attached an Internet Gateway to your VPC, but an EC2 instance inside still can't reach the internet. What 3 configurations should you check?
- 2
- 3 >_ What is the functional difference between a Security Group and a Network ACL? Which one is stateful?
- 3
- 4 >_ If databases must be strictly kept in a private subnet, how can they securely download critical OS patches?
- 5
- 6 >_ Can a VPC span multiple Regions? Can it span multiple AZs?

The Forward-Deployed Mindset

Why VPCs are the invisible glue of modern engineering.

